USN [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [22TCSDE22]

FIFTH SEMESTER B.Tech EXAMINATION, NOVEMBER-2024

DATA ANALYTICS AND MACHINE LEARNING FOR CYBER SECURITY

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5x2=10**		
1.	**a.**	Explain why data preprocessing is necessary in data science?	Understanding	CO2	**02**
	b.	What is NoSQL and when is it used?	Understanding	CO1	**02**
	c.	What is the difference between structured, and semi-structured data?	Understanding	CO1	**02**
	d.	Discuss the importance of feature selection and how it improves the performance of machine learning models.	Understanding	CO2	**02**
	e.	What is the main difference between supervised and unsupervised learning?	Understanding	CO3	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain the concept of Gradient Descent in machine learning.	Understanding	CO2	**05**
		OR			
	b.	Describe the types of gradient descent and their differences.	Understanding	CO2	**05**
3.	**a.**	Explain the role of hyperparameters in machine learning models	Understanding	CO1	**05**
		OR			
	b.	Differentiate between first-order and second-order optimization methods.	Understanding	CO1	**05**
4.	**a.**	Explain the concept of outlier detection in data cleaning?	Understanding	CO1	**05**
		OR			
	b.	How does feature extraction help in data reduction?	Understanding	CO2	**05**
5.	**a.**	Describe the process of handling missing data in datasets and its impact on data analysis.	Applying	CO5	**05**
		OR			
	b.	What are the common methods used to prevent overfitting in deep learning models?	Understanding	CO1	**05**
6.	**a.**	Explain the working principle of a perceptron.	Understanding	CO3	**05**
		OR			
	b.	What is information gain, and how is it used in decision trees?	Understanding	CO3	**05**

7.	**a.**	Explain how logistic regression is used for binary classification problems.	Applying	CO4	**05**
		OR			
	b.	Describe the working of the K-means algorithm and discuss how the elbow method helps in selecting the optimal number of clusters.	Understanding	CO3	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Describe the purpose of backpropagation in neural networks.	Applying	CO6	**10**
		OR			
	b.	Explain the limitations of traditional ANNs for image processing and how CNNs overcome these limitations.	Applying	CO6	**10**
9.	**a.**	What is the role of a hyperplane in a Support Vector Machine?	Understanding	CO3	**10**
		OR			
	b.	How does a support vector machine (SVM) maximize the margin between different classes?"	Understanding	CO3	**10**
10.	**a.**	What is the purpose of Principal Component Analysis (PCA) in machine learning?	Applying	CO4	**10**
		OR			
	b.	Describe the steps involved in building a decision tree using the CART algorithm.	Understanding	CO3	**10**
11.	**a.**	Explain the concept of probability mass function (PMF) with an example.	Understanding	CO2	**10**
		OR			
	b.	Design a deep learning model to handle the phishing attack.	Applying	CO6	**10**
12.	**a.**	Define Bayes' Theorem and explain its significance in probability theory?	Understanding	CO2	**10**
		OR			
	b.	Compare Bayesian Optimization with Grid Search for hyper parameter tuning	Applying	CO6	**10**
13.	**a.**	Differentiate between precision and recall.	Understanding	CO2	**10**
		OR			
	b.	Explain how anomaly detection works in unsupervised learning.	Understanding	CO3	**10**